

Qualitative Community Coding of the Literature Selection

Basis: 250 records from the uploaded CSV. Only the columns **title** and **keywords** were used for the thematic coding, as requested. The categories below represent dominant scientific communities visible in the literature selection. They are interpretive and content-based; some papers may also belong to secondary communities.

Overview of Dominant Communities

Community category	N	Share
Assessment, Measurement, Psychometrics, and Program Evaluation	48	19.2%
STEM, Engineering, Mathematics, and Life Sciences Education	35	14.0%
AI, Learning Analytics, NLP, and Educational Data Science	23	9.2%
Higher Education, Graduate Education, and Student Success	21	8.4%
Motivation, Self-Regulation, Wellbeing, and Socio-Emotional Development	20	8.0%
Digital, Online, VR/AR, and Technology-Enhanced Learning	18	7.2%
Teacher Education, Professional Development, and Instructional Practice	18	7.2%
Language, Literacy, Reading, and Multilingual Education	16	6.4%
Equity, Inclusion, Diversity, and Special/Gifted Education	14	5.6%
Other / Broad Educational Research	10	4.0%
Research Methods, Bibliometrics, Reviews, and Open Science	10	4.0%
Early Childhood, Family, Home Learning, and Developmental Psychology	8	3.2%
Curriculum, Disciplinary Pedagogy, and Learning Design	4	1.6%
Educational Policy, Systems, Absenteeism, and Institutional Governance	3	1.2%
Science Communication, Informal Learning, Museums, and Outreach	2	0.8%

Interpretation

The selection is strongest in STEM/disciplinary education, assessment and psychometrics, AI/data-science approaches, digital learning, teacher education, and higher education research. Equity/inclusion, wellbeing/motivation, language/literacy, family/developmental research, policy, open science, and informal science communication appear as smaller but clearly identifiable communities. Because the coding relies on titles and keywords only, categories should be understood as a high-level map rather than a full-text classification.

Category Definitions and Examples

Assessment, Measurement, Psychometrics, and Program Evaluation (n=48; 19.2%)

Definition	Communities focused on instruments, tests, validation, psychometrics, learning outcomes, scale construction, educational measurement, comparative judgment, and evaluation designs.
Example	<i>Development and validation of a rapid and precise online sentence reading efficiency assessment</i> Keywords: dyslexia; reading fluency assessment; screening tools; reading fluency and comprehension; progress monitoring; psychometrics

STEM, Engineering, Mathematics, and Life Sciences Education (n=35; 14.0%)

Definition	Research communities centred on teaching and learning in STEM fields, including mathematics, engineering, life sciences, bioinformatics, computer science education, spatial reasoning, and domain-specific disciplinary learning.
Example	<i>Nuclear history, politics, and futures from (A)toms-to(Z)oom: Design and deployment of a remote-learning special-topics course for nuclear engineering education</i> Keywords: nuclear physics and engineering history; remote education; STEM pedagogy; nontechnical engineering education; history of science

AI, Learning Analytics, NLP, and Educational Data Science (n=23; 9.2%)

Definition	Work using AI, machine learning, large language models, NLP, predictive modelling, learning analytics, automated feedback or assessment, text mining, and educational data science.
Example	<i>Explainable Automated Essay Scoring: Deep Learning Really Has Pedagogical Value</i> Keywords: explainable artificial intelligence; SHAP; automated essay scoring; deep learning; trust; learning analytics; feedback; rubric

Higher Education, Graduate Education, and Student Success (n=21; 8.4%)

Definition	Research communities concerned with universities, college access and persistence, graduate/doctoral education, student success, advisor relationships, career outcomes, and tertiary/postsecondary transitions.
Example	<i>Making strides in doctoral-level career outcomes reporting: a review of classification and visualization methodologies in graduate education</i> Keywords: graduate education; doctoral training; program evaluation; career outcomes; taxonomy; PhD

Motivation, Self-Regulation, Wellbeing, and Socio-Emotional Development (n=20; 8.0%)

Definition	Communities examining motivation, engagement, self-efficacy, self-regulation, belonging, wellbeing, mental health, stress, identity, attitudes, anxiety, help-seeking, and socio-emotional outcomes.
Example	<i>Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic: Emotional Stability, Mental Health, and Household Resources</i> Keywords: COVID-19 pandemic, college students, psychological well-being, emotions, experience sampling method, personality, intraindividual trajectories

Digital, Online, VR/AR, and Technology-Enhanced Learning (n=18; 7.2%)

Definition	Communities studying digital learning environments, online and distance education, educational technology, virtual or augmented reality, e-learning, learning apps, games, CSCL, and technology-enhanced instruction.
Example	<i>The limited role of expectancy-value beliefs, self-efficacy, and perceived attentional control in predicting online learning outcomes in a general education course</i> Keywords: academic motivation; self-efficacy; expectancy-value; higher education; introductory psychology; online courses

Teacher Education, Professional Development, and Instructional Practice (n=18; 7.2%)

Definition	Communities studying teachers, teacher educators, school leaders, professional learning, instructional practice, implementation of evidence-informed teaching, and educator wellbeing or work conditions.
-------------------	---

Example	<i>Mind the Gap: How Teachers' Professional Development Preferences Relate to Scientific Recommendations</i> Keywords: teacher professional development; video vignettes; evidence-based interventions; teacher preferences; implementation
----------------	--

Language, Literacy, Reading, and Multilingual Education (n=16; 6.4%)

Definition	Communities studying language learning, reading, literacy, dyslexia, vocabulary, bilingualism, second-language learning, oral reading fluency, speech perception, textbooks, and literacy-integrated content learning.
Example	<i>Orthographic similarity ratings for English-Spanish cognates from the academic word list</i> Keywords: cognates; academic word list; orthographic similarity; reading; education; bilingualism

Equity, Inclusion, Diversity, and Special/Gifted Education (n=14; 5.6%)

Definition	Communities addressing educational equity, inclusion, representation, disability, gifted education, race/ethnicity, gender, socioeconomic inequalities, first-generation status, rurality, and social justice.
Example	<i>Effect of Local Norms on Racial and Ethnic Representation in Gifted Education</i> Keywords: assessment; descriptive analysis; diversity; equity; gifted education; hierarchical linear modeling; policy analysis

Other / Broad Educational Research (n=10; 4.0%)

Definition	Papers whose title and keywords did not clearly signal one of the recurring communities above, or that combine general educational-psychology themes without a distinctive community label.
Example	<i>Improving Comprehension: Intelligent Tutoring System Explaining the Domain Rules When Students Break Them</i> Keywords: Blooms taxonomy; adaptive learning; cognitive learning; constraint-based tutors; intelligent tutoring system

Research Methods, Bibliometrics, Reviews, and Open Science (n=10; 4.0%)

Definition	Meta-research and methodological communities concerned with reviews, bibliometrics, science mapping, open science, data sharing, preregistration, replication, transparency, statistical methods, and research methodology.
Example	<i>Open Education Science</i> Keywords: preregistration; open science; registered report; open access

Early Childhood, Family, Home Learning, and Developmental Psychology (n=8; 3.2%)

Definition	Research communities focused on younger learners, parent/family involvement, home learning environments, early development, preschool/elementary children, adolescent development, and developmental psychology.
Example	<i>Relations between the Home Learning Environment and the Literacy and Mathematics Skills of Eight-Year-Old Canadian Children</i> Keywords: home literacy; home numeracy; early academic skills; grade 3 children; parental involvement; home mathematics environment; home activities

Curriculum, Disciplinary Pedagogy, and Learning Design (n=4; 1.6%)

Definition	Communities around curriculum design, course design, disciplinary pedagogy, project-based learning, concept mapping, learning transfer, learning materials, and instructional design approaches.
Example	<i>Empowering Future HR Professionals: A Design-Based Research Approach to Project-Based Learning in Work and Organizational Psychology</i> Keywords: project-based learning; open pedagogy; higher education development; human resource development; design-based research; student-centered learning; psychology education; openness; academic culture transformation

Educational Policy, Systems, Absenteeism, and Institutional Governance (n=3; 1.2%)

Definition	Communities analysing policy, administrative or system-level data, absenteeism, school resources, regional differences, governance, policy thresholds, and institutional conditions.
Example	<i>Academic Harms of Missing High School and the Accuracy of Current Policy Thresholds: Analysis of Preregistered Administrative Data From a California School District</i> Keywords: absenteeism; high school; preregistration; secondary data

Science Communication, Informal Learning, Museums, and Outreach (n=2; 0.8%)

Definition	Communities studying learning beyond formal classrooms, science communication, outreach, museums, public sentiment, online affinity spaces, researcher visibility, and academic social networks.
Example	<i>Characterizing an online, science-based affinity space using topic modelling, diversity indices, and social network analysis</i> Keywords: Machine learning; science communication; science outreach; InfoMaps; Netlytic; NodeXL

Coding Procedure

The coding was conducted inductively from repeated terms and topic signals in the title and keyword fields. Terms such as STEM, assessment, learning analytics, teacher education, inclusive education, motivation, language, home learning, policy, bibliometrics, and science communication were used as anchors for scientific communities. For the count table, each record was assigned to one dominant category based on the strongest title/keyword signal. This simplifies overlapping literatures into a readable community map.

Suggested use: treat the categories as an analytic codebook for a first-cycle qualitative coding pass. If the next step is publication-level coding, the categories can be refined through double coding, full-text inspection, and allowing multiple labels per paper.